Activity 3.2: Noise and Speech Activity Analysis

Assignee: Humesh Sharma **Work Item:** Story **Allocated Duration:** 1h 30m **Status:** In Progress

1. Overview & Scope

This activity focuses on analyzing the extracted audio signals to identify ambient noise levels and speech activity patterns. Audio features such as energy levels, amplitude variation, and short-time signal characteristics were computed for each segment. These features were used to differentiate between periods of silence, background noise, and active speech.

Noise level trends provide insights into classroom environment conditions, while speech activity detection indicates interaction intensity and participation frequency. The results of this analysis were aggregated over time to produce session-level audio context metrics.

The generated audio insights were structured in a format suitable for fusion with visual engagement data, enabling the multimodal agent to reason about overall session dynamics.

2. Acoustic Analysis & Feature Specifications

Analyzed Feature	Classification Target	Functional Role in Session Analysis
Energy & Amplitude Variation	Silence vs. Active Signal	Computes short-time signal energy across segmented windows to separate quiet intervals from acoustic events.
Ambient Noise Trends	Background Noise Floor	Evaluates baseline classroom environment conditions and identifies persistent non-speech acoustic disturbances.
Speech Activity Detection	Active Speech Patterns	Measures classroom interaction intensity, vocal engagement, and student/instructor participation frequency.
Temporal Aggregation	Session-Level Metrics	Structures time-series audio context insights for downstream fusion with visual engagement indicators.

3. Client Capture & Streaming (`frontend/src/components/AudioCapture.jsx`)

```
frontend/src/components/AudioCapture.jsx                    React / WebSocket Audio Client
```

```
import React, { useEffect, useRef, useState }  ;

  = ({ active, onMetrics }) => {
     [isRecording, setIsRecording] = ();
     wsRef = ();
     audioContextRef = ();
     processorRef = ();
     inputRef = ();

    (() => {
         (active) {
            ();
        }  {
            ();
        }
         () => ();
    }, [active]);

      =  () => {
         {
             stream =  navigator.mediaDevices.({ audio:  });
            ();


             ws =  ();
            wsRef.current =
ws;from'react'constAudioCaptureconstuseStatefalseconstuseRefnullconstuseRefnullconstuseRefnu
llconstuseRefnulluseEffectifstartCaptureelsestopCapturereturnstopCaptureconststartCaptureasy
nctryconstawaitgetUserMediatruesetIsRecordingtrue// Connect
WSconstnewWebSocket'ws://localhost:8000/ws/audio'
```

4. Multimodal Fusion & Context Integration

> **Multimodal Fusion Note:** Real-time audio streams captured via the client WebSocket interface (`ws://localhost:8000/ws/audio`) undergo short-time feature extraction to output structured noise and speech activity metrics. These session-level audio insights are fused with visual engagement data so the multimodal agent can reason comprehensively about classroom dynamics.